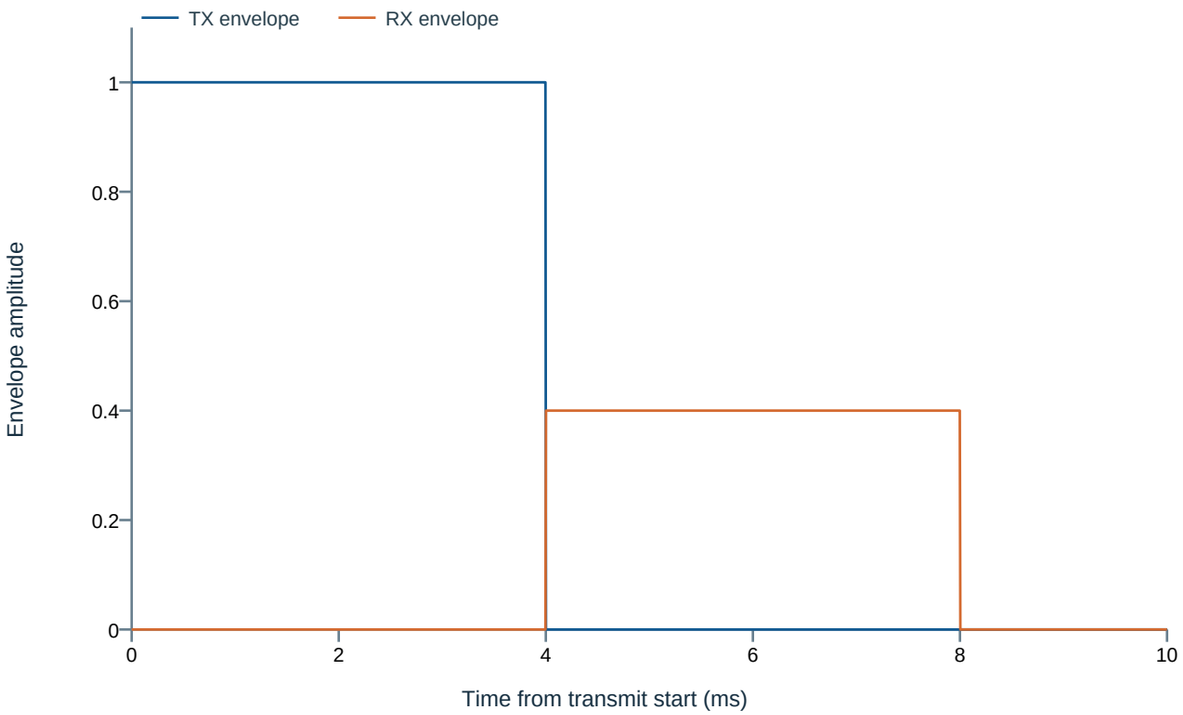
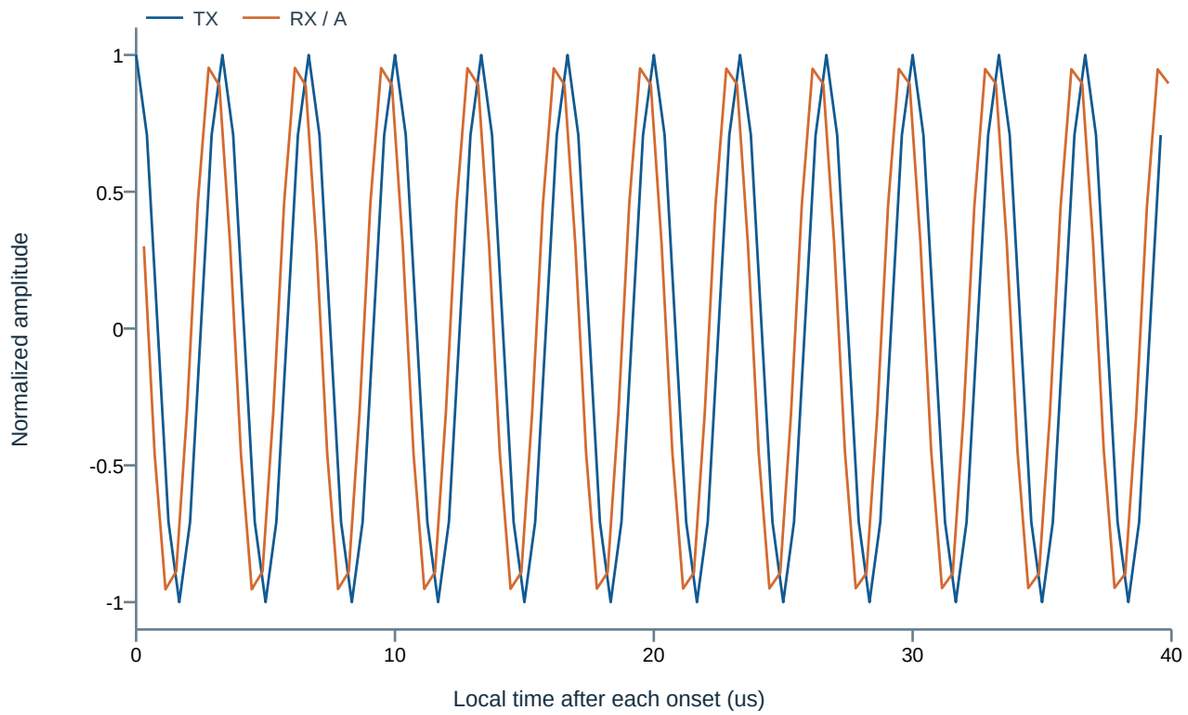
1 / Transmit and receive waveforms

Single point scatterer: c=1500 m/s, f0=300 kHz, Tp=4 ms, R0=3 m, v=+0.2 m/s

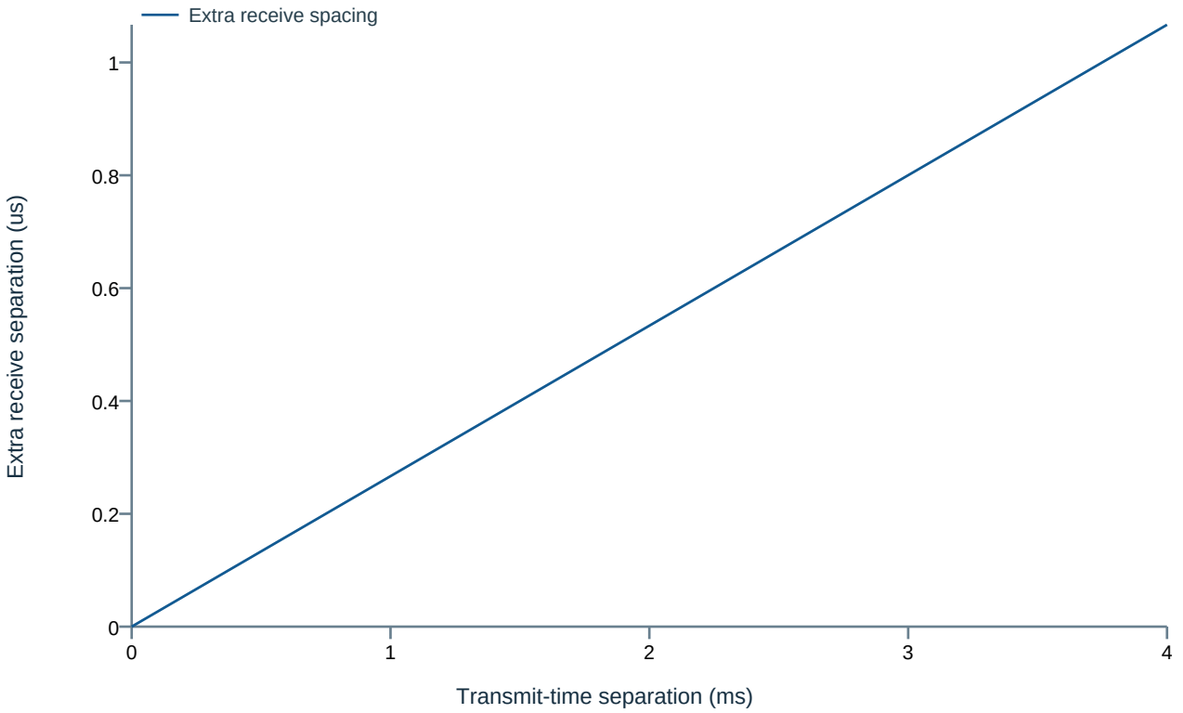
A. Pulse envelopes locate TX and RX



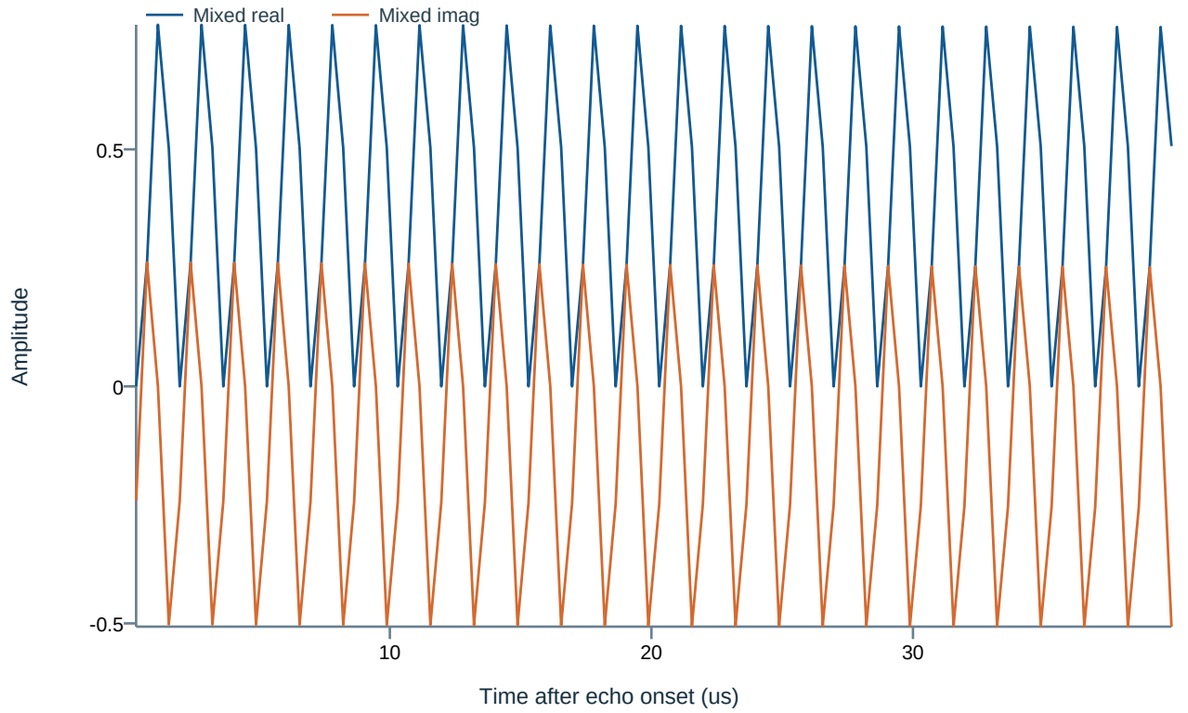
B. First 40 us: many carrier cycles



C. Time stretching (onset removed)



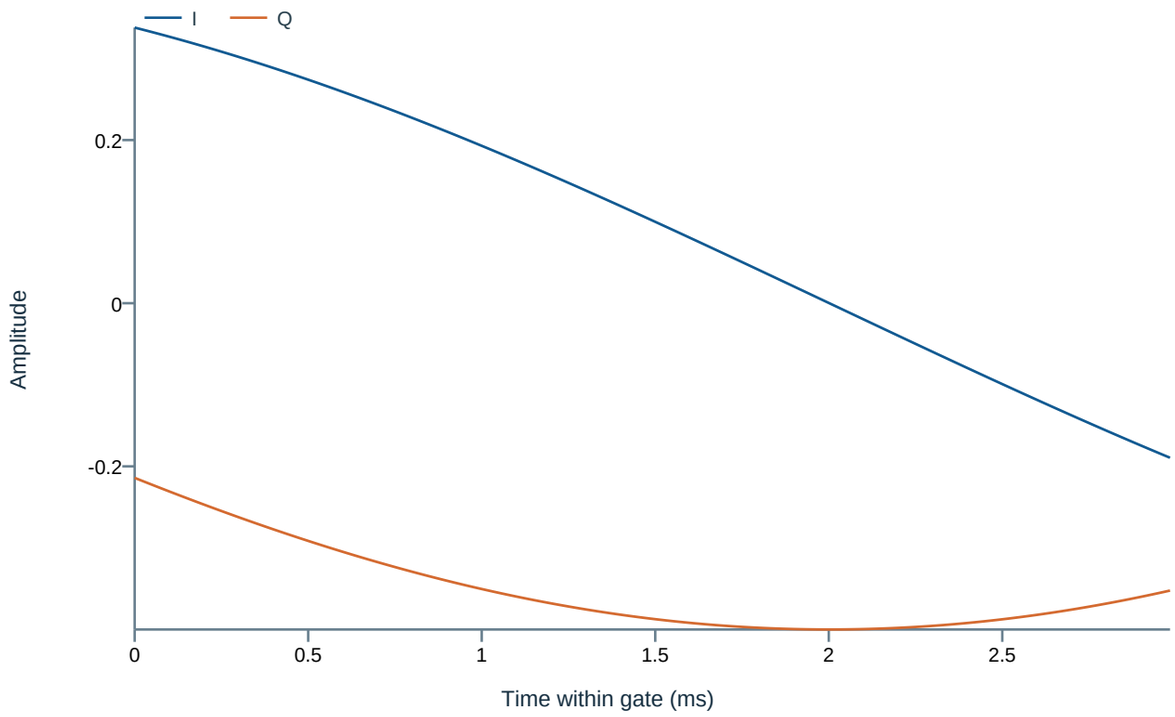
D. Mixer output before low-pass



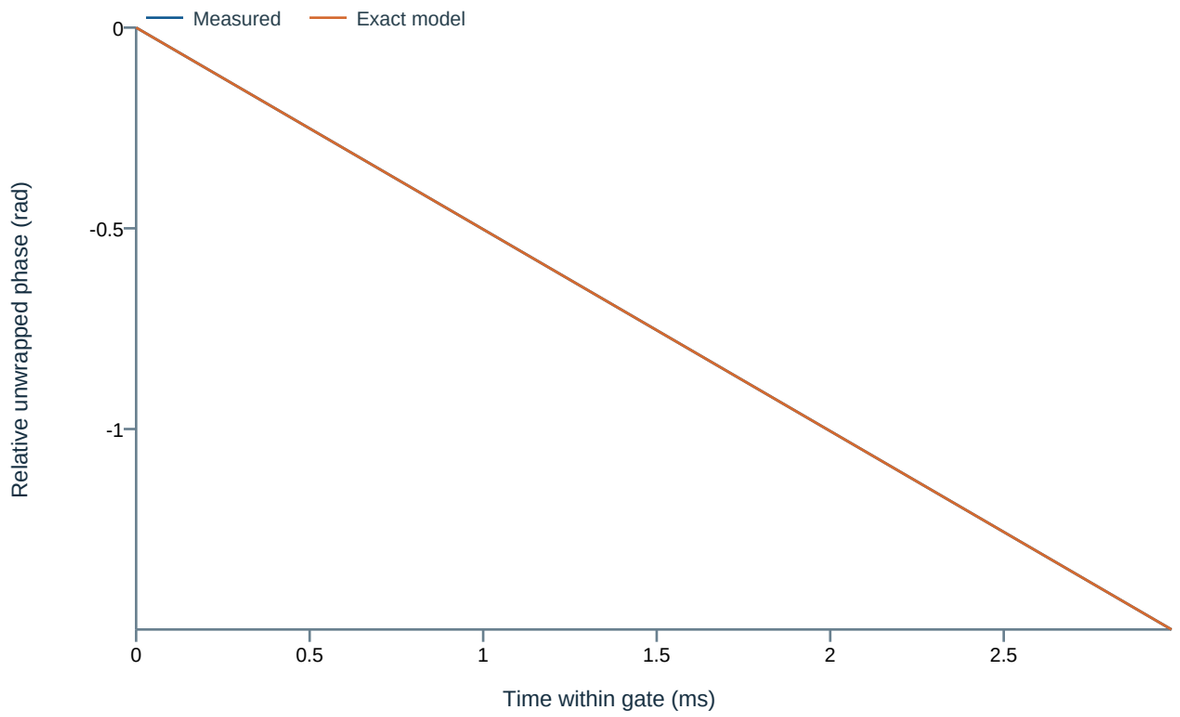
2 / I/Q, phase and lag correlation

I/Q sample rate = 60 kHz | Gate = 180 samples (3 ms) | Lag = 30 samples (0.5 ms) | 150 products

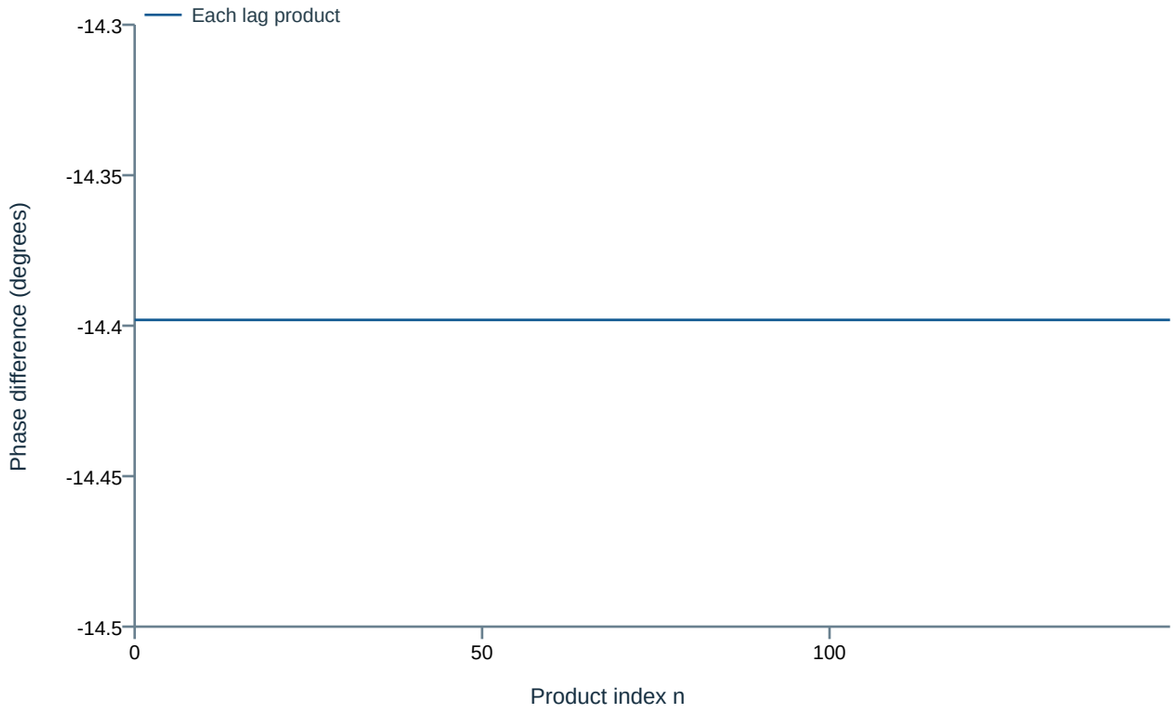
A. Filtered I/Q inside selected echo gate



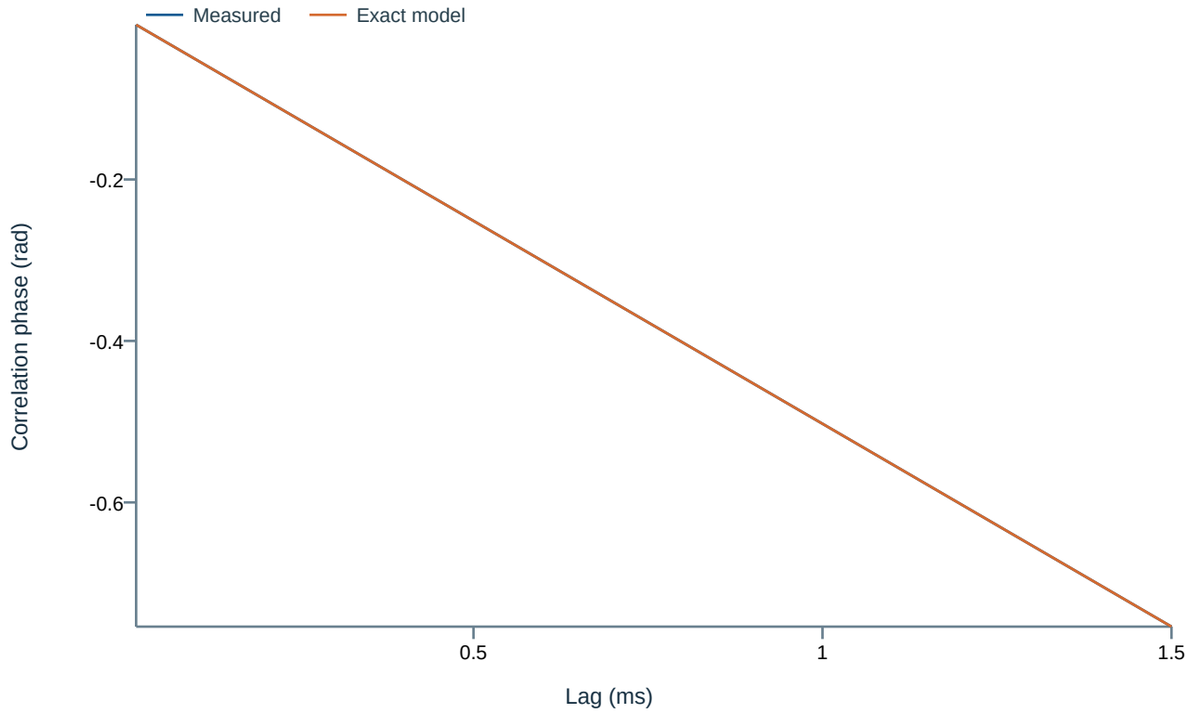
B. Phase accumulates at the Doppler rate



C. Every product compares samples 30 apart



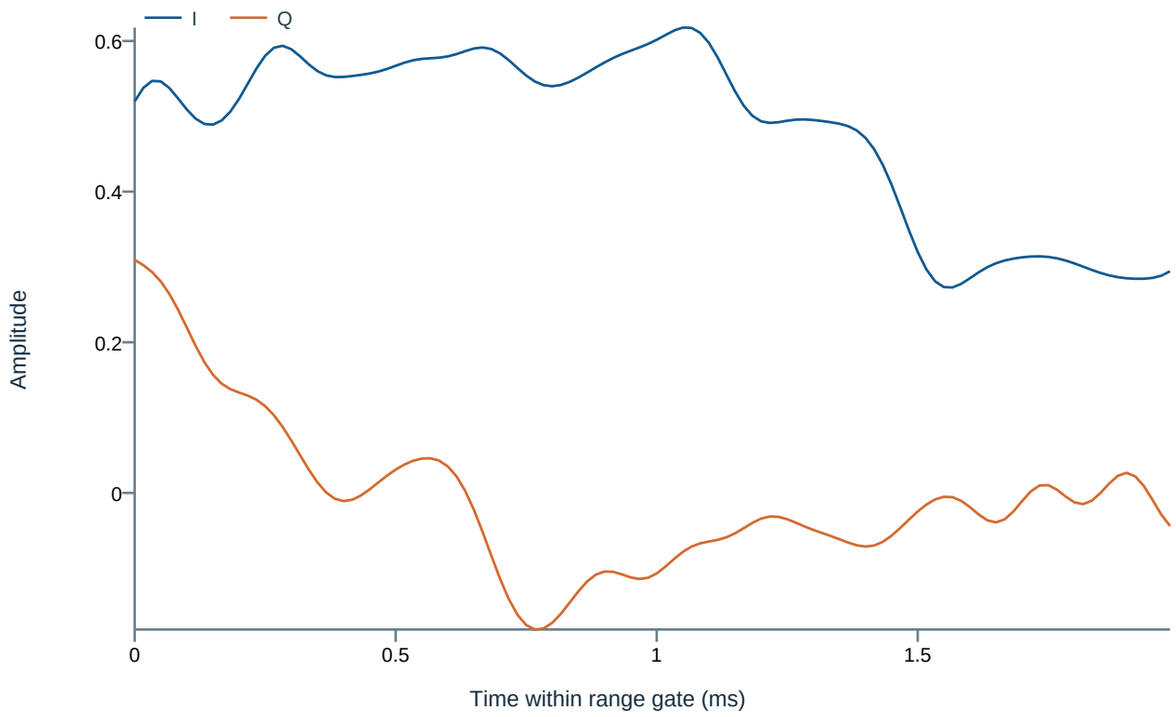
D. Different lag, same constant Doppler



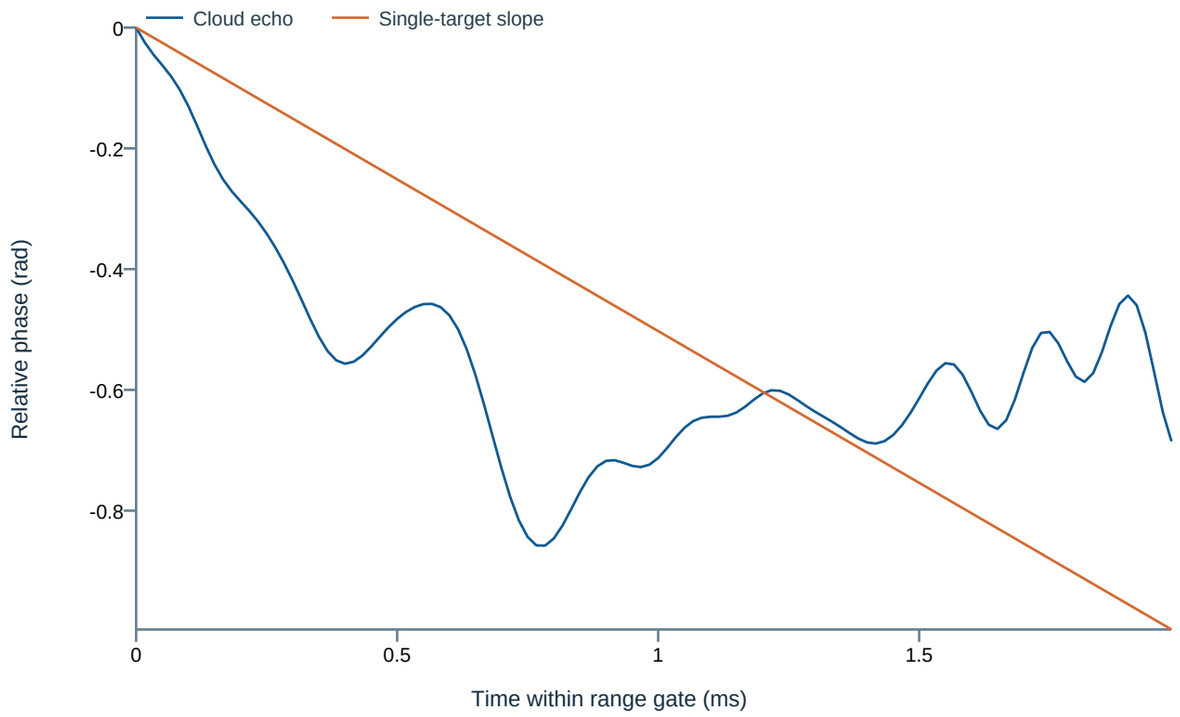
3 / A cloud of scatterers changes the error

600 moving scatterers | Same constant true v=+0.2 m/s | 2 ms receive gate | No added receiver noise

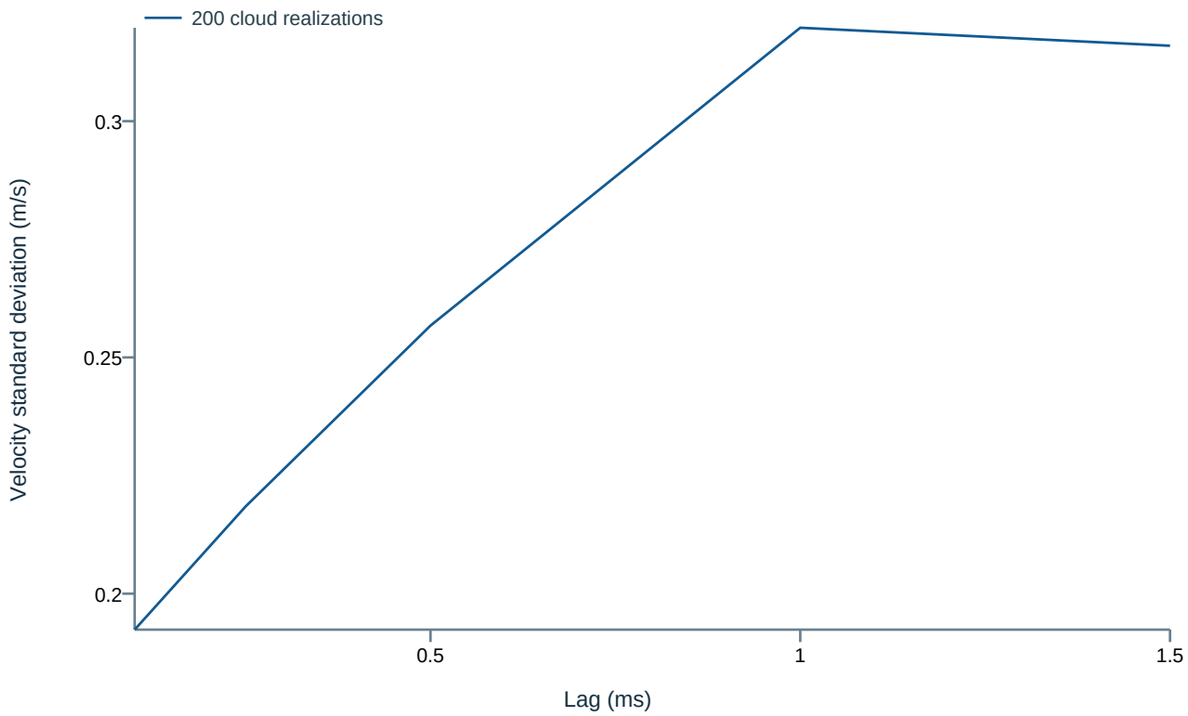
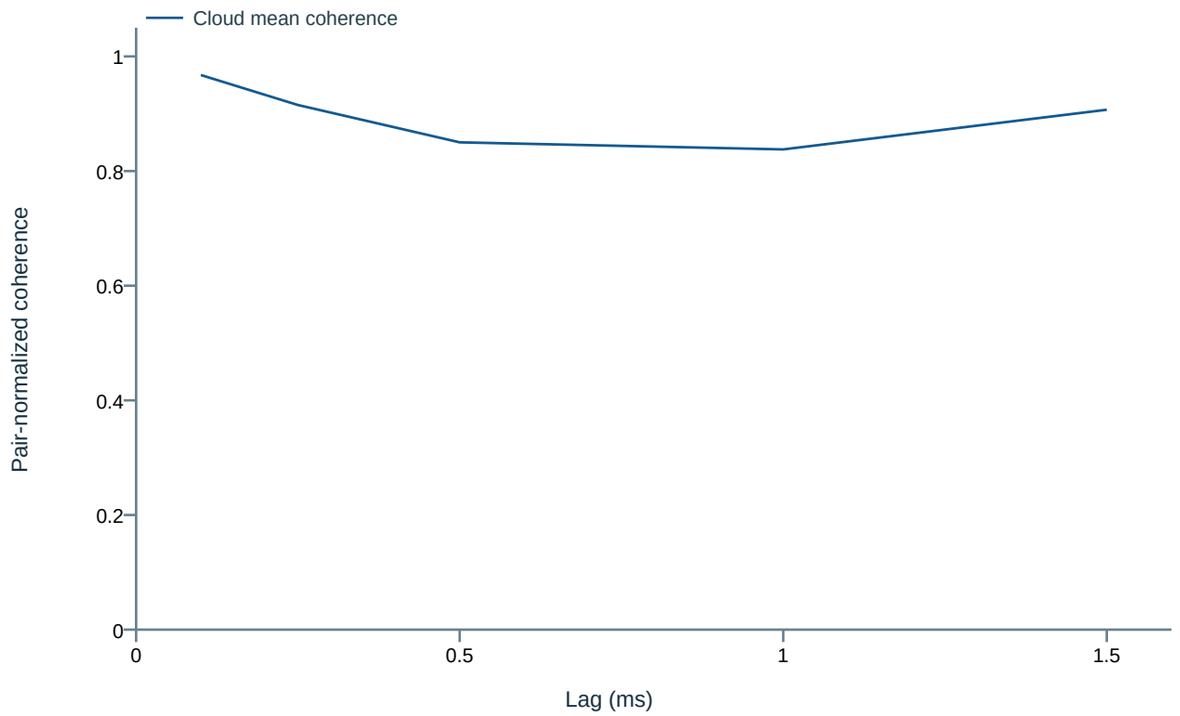
A. Echo is a changing sum of many phasors



B. Scatterer turnover adds phase fluctuations



C. Measured single-ping random spread



D. Fixed window: coherence depends on lag